

Project Design Phase-II: Technology Stack & Technical Architecture

Project Overview

- **Project Name:** Heart Disease Analysis[cite: 28]
- **Date:** 8 July 2026[cite: 28]
- **Maximum Marks:** 4 Marks[cite: 28]
- **Team Leader:** Mounika Sanapathi[cite: 28]
- **Team Members:** Thakasi Balaji, Kale Sai Praneeth, Navyasri Gonnuri, Shanthi Kumari Challa[cite: 28]

Technical Architecture

The system architecture tracks patient health records from ingestion through automated preparation up to secure interactive deployment for clinical review.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface[cite: 28]	Enables medical professionals to interact with clinical metrics and view risk dashboards via standard browsers[cite: 28]. Features responsive diagnostic embeds[cite: 28].	HTML5, CSS3, JavaScript[cite: 28]
2.	Application Logic-1[cite: 28]	Manages clinical web platform routing, coordinates secure page rendering, and nests interactive Tableau embeds[cite: 28].	Python, Flask[cite: 28]

3.	Application Logic-2[cite: 28]	Standardizes raw health data anomalies, fills missing biometric features, and normalizes values before visualization[cite: 28].	Python[cite: 28]
4.	Application Logic-3[cite: 28]	Seamlessly injects the global medical dashboard views and guided user stories into the secure patient data portal web pages[cite: 28].	Tableau Public (Embedded View), HTML5[cite: 28]
5.	Database[cite: 28]	This clinical project omits standalone database dependencies; health data loads directly from source record extracts[cite: 28].	Not Applicable for this project[cite: 28]
6.	Cloud Database[cite: 28]	Remote managed enterprise databases are omitted from the project's current deployment configuration scope[cite: 28].	Not Applicable for this project[cite: 28]
7.	File Storage[cite: 28]	Standard physical repository storage of structured medical datasets containing demographic and diagnostic values[cite: 28].	Local File System (CSV)[cite: 28]
8.	External API-1[cite: 28]	No third-party clinical or transactional external API interfaces are utilized in this system layer[cite: 28].	Not Applicable for this project[cite: 28]
9.	External API-2[cite: 28]	Auxiliary outward communication connections or services are completely omitted from the present build[cite: 28].	Not Applicable for this project[cite: 28]
10.	Machine Learning Model[cite: 28]	Predictive predictive modeling nodes or downstream automated classification blocks are omitted[cite: 28].	Not Applicable for this project[cite: 28]
11.	Infrastructure (Server / Cloud) [cite: 28]	Runs local secure workstation sessions. Flask instances deploy across standard developer setups	Local Machine, Tableau Public

		utilizing Git tracking via GitHub[cite: 28].	(Dashboard Hosting)[cite: 28]
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Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks[cite: 28]	Built on reliable open-source components to maintain an agile, transparent, and reproducible codebase[cite: 28].	Flask (Python), HTML5, CSS3, JavaScript[cite: 28]
2.	Security Implementations[cite: 28]	Published views run via Tableau Public with access managed safely over GitHub; no sensitive, identifiable patient data is exposed[cite: 28].	GitHub repository access control[cite: 28]
3.	Scalable Architecture[cite: 28]	Modular interface design allows clinicians to add new health parameters and sub-dashboards without rewriting base routing code[cite: 28].	Modular Flask application structure[cite: 28]
4.	Availability[cite: 28]	Visual clinical tools remain accessible inside the web portal using decentralized visual presentation hosting servers[cite: 28].	Tableau Public hosting[cite: 28]
5.	Performance[cite: 28]	Fast diagnostic chart rendering achieved through heavily filtered data extracts and pre-aggregated biometric views[cite: 28].	Tableau Public rendering optimization[cite: 28]

Prepared as part of the Heart Disease Analysis and Patient Risk Trends capstone project.[cite:
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